Band-1 Hadamard Order 3044 — Miyamoto (H_760 Paley-I×2); Miyamoto-Only Bonus; Gap 2052 Still OPEN

H_380/H_760/C_762/H_3044 PASS (Miyamoto); 3044 NOT in SPK (Paley-I/II/Kronecker FAIL); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_2836_Miyamoto`; `2026-09-15_Band1_Hadamard_2804_Miyamoto`; `2026-09-14_Band1_Hadamard_2276_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_3044_miyamoto/`.
Cutoff. 2026-09-15 (~03:09:43Z; ~11:09PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 380, 760, and 3044 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-I seed H_{380} Sylvester-doubled to H_{760}. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 3044 is a **Miyamoto-only bonus** (**not** in SPK): honest SPK audit — Paley-I **FAIL** ($q = 3043$ not prime); Paley-II **FAIL** ($q = 1521$ not prime); Kronecker **FAIL** ($3044 = 4 \cdot 761$, 761 not Hadamard-admissible). Paley-II path for seed H_{760} **REJECTED** ($760 = 2(q+1) \Rightarrow q = 379 \equiv 3 \pmod 4$). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 2836 on box. This increment constructs the **next preferred** Miyamoto-eligible target after $q = 709$: **order 3044** ($3044 = 4 \cdot 761$, 761 prime $\equiv 1 \pmod 4$). Seed H_{380} via Paley type I ($q = 379$ prime $\equiv 3 \pmod 4$; order $q+1 = 380$), then Sylvester double H_{760} (Paley-II path for 760 **rejected**). Symmetric Paley conference C_{762} over prime 761. $H_{3044} = \text{Miyamoto}(H_{760}, C_{762})$. Unit tests **28/28 PASS**. Order 3044 does **not** enter SPK (Miyamoto-only bonus). SPK next gap remains **2052** (OPEN). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 3044; Miyamoto; Paley type I; Sylvester doubling; Paley conference; Miyamoto-only bonus; constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{380} constructed and verified (Paley type I, $q = 379$). 2. H_{760} constructed and verified (Sylvester double of H_{380}). Paley-II path **rejected**. 3. C_{762} constructed and verified (Paley conference, $q = 761$): $CC^\top = 761I$, zero diagonal. 4. H_{3044} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. SPK: $3044 \notin$ SPK (Miyamoto-only bonus). Paley-I, Paley-II, and Kronecker paths for 3044 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$3044 = 4 \cdot 761$
$q = 761$	prime, $761 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{760}
Needed conference	order $762 = q + 1$
H_{380} path	Paley type I, $q = 379$ (prime $\equiv 3 \pmod 4$); $q + 1 = 380$
H_{760} path (used)	Sylvester double of H_{380}
Paley-II for H_{760}	**REJECTED**: $760 = 2(q + 1) \Rightarrow q = 379 \equiv 3 \pmod 4$ (needs $\equiv 1$)
Conference path	Symmetric Paley over prime 761
Paley-I for 3044	**REJECTED**: $q' = 3043$ not prime
Paley-II for 3044	**REJECTED**: $q' = 1521$ not prime ($1521 = 39^2$)

Quantity	Value
Kronecker for 3044	**REJECTED**: only 4×761; $761 \not\equiv 0 \pmod 4$ (not admissible)
SPK membership	**False** (Miyamoto-only bonus)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{380}	Paley type I, $q = 379$	0	0.0918	**PASS**
H_{760}	Sylvester double of H_{380}	0	0.1987	**PASS**
C_{762}	Paley conference, $q = 761$	0	0.4911	**PASS**
H_{3044}	Miyamoto(H_{760}, C_{762})	0	12.1394	**PASS**

Primary Miyamoto pipeline wall (incl. unit tests + SPK refresh) $\approx$ 13.441 s. Compressed ± 1 matrices saved as `H380_pm1.npz`, `H760_pm1.npz`, `C762_pm1.npz`, `H3044_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior ships (incl. 708, 2836, 700, 2804) $\cup$ verified H_{380}, H_{760} $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 3044 therefore remains **outside** SPK.

Quantity	Value
Known count	883
3044 in SPK	**False** (Miyamoto-only bonus)
760 in SPK	True (Sylvester double of Paley-I H_{380})
380 in SPK	True (Paley I $q = 379$)
2836 in SPK	False (prior Miyamoto bonus)
2804 in SPK	True (prior Paley I)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, 2148, 2156, 2172, 2212, 2284…

5. Gap 2052 (still OPEN; non-finding ONLY)

This ship does **not** fill gap 2052. Prior triage stands: Paley-I/II / Kronecker / Miyamoto-Paley-conf all **FAIL** for order 2052. Claim level for 2052 remains method triage / non-finding ONLY. Non-finding $\neq$ counterexample; Hadamard conjecture remains OPEN. No H_{2052} matrix is claimed.

6. Unit tests (28/28)

Test	Result
H4_sylvester	PASS
H20_paley_I_control	PASS
paley_II_q5_H12	PASS
paley_II_q13_H28	PASS
miyamoto_H20_from_H4_conf6	PASS
paley_I_q7_H8_double_H16	PASS
paley_I_q11_H12	PASS
q379_is_prime	PASS
q379_equiv_3_mod_4	PASS
H380_order_formula	PASS
H760_double_formula	PASS
q761_is_prime	PASS
q761_equiv_1_mod_4	PASS
n3044_equals_4_times_761	PASS
H760_order_formula	PASS
C762_order_formula	PASS
paley_II_H760_path_rejected	PASS
paley_I_H3044_path_rejected	PASS
paley_II_H3044_path_rejected	PASS
kronecker_H3044_path_rejected	PASS
q3043_not_prime	PASS
q1521_not_prime	PASS
reject_all_ones_H8	PASS
conference_6_paley_control	PASS
miyamoto_eligible_3044_arith	PASS
conference_762_shape_expected	PASS
gap_2052_not_filled_by_this_ship	PASS
3044_miyamoto_only_bonus_arith	PASS

7. Next optional targets

Miyamoto-eligible primes $q > 761$, $q \equiv 1 \pmod 4$ (sample):

q	$n = 4q$	Need H_{q-1}	On-box path notes
769	3076	768	Double(PaleyI half=384), Kronecker(4x192)
773	3092	772	—
797	3188	796	PaleyII(q=397)

q	$n = 4q$	Need H_{q-1}	On-box path notes
809	3236	808	Kronecker(404x2)
821	3284	820	PaleyII(q=409)
829	3316	828	PaleyI(q=827)
853	3412	852	—
857	3428	856	Kronecker(428x2)

Preferred next. Method-attack remaining SPK gap **2052** (GS/SDS for order ~513 if literature seeds appear), **or** construct Miyamoto $q = 769 \to n = 3076$ (Double/Kronecker notes), **or** $q = 797 \to n = 3188$ (Paley-II $q = 397$), **or** Track B privacy / Object 1 meter await. GATE-04 remains LOCKED.

8. Non-claims (explicit)

- Not the full Hadamard conjecture.
- Not Clay-closed / finished-ToE / RH / P≠NP / YM / Frankl / NS-solved / AC-solved.
- Gap **2052** remains **OPEN**.
- Order 3044 is **not** in SPK (Miyamoto-only bonus).
- GATE-04 collection **LOCKED**; no invented J.
- `clay_closed = false`.

9. Artifacts

Lab: `/workspace/experiments/band1/hadamard_3044_miyamoto/`
Verifier: `verify_hadamard_3044.py`
Results: `results.json`, `literature_ledger.json`, `SUMMARY.txt`
Matrices: `H380_pm1.npz`, `H760_pm1.npz`, `C762_pm1.npz`, `H3044_pm1.npz`
Shelf: `/workspace/DOWNLOAD_PDFs/2026-09-15_Band1_Hadamard_3044_Miyamoto.pdf`